



KBPCOES	COURSE SYLLABUS	DOC NO.: KBPCOES/ACD/F/05-00-04
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Program / Department: Computer Science & Engineering	Class: TY	Semester: VI	Div: Diamond
Course: Compiler Design	Course Code: BTCOC601	Course Coordinator: Mrs.Bhokare U.M.	

BTCOC601: Compiler Design

[Unit 1] Introduction to Compiling [7 Hours]

Definition, analysis of the source program, the phases of a compiler, the grouping of phases, Compiler Construction tools, A simple one-pass compiler.

[Unit 2] Lexical Analysis [7 Hours]

The role of the Lexical analyzer, Input buffering, Specification of Tokens, A Language for Specifying Lexical Analyzers, Design of a Lexical Analyzer generator.

[Unit 3] Syntax Analysis [7 Hours]

The role of the Parser, Context-free grammars, Writing a Grammar, Top-Down Parsing, Bottom-Up Parsing, Operator-precedence Parsing, LR Parsers, Using Ambiguous Grammars, Parser Generators.

[Unit 4] Syntax-Directed Translation [7 Hours]

Definitions, Construction of Syntax Trees, Bottom-Up Evaluation of S- Attributed definitions, Top-Down Translation, Bottom-Up Evaluation of Inherited attributes. Intermediate Languages, Declarations, Assignment Statements, Boolean Expressions, Case Statements, Back patching, Procedure Calls.

[Unit 5] Code Generation [7 Hours]

Issues in the Design of a Code Generator, The target Machine, Run-Time Storage Management, Basic Blocks and Flow Graphs, Next-Use Information, Simple Code Generator, Register allocation and Assignment, The DAG Representation of Basic Blocks, Generating Code from DAGs, Dynamic Programming, Code- Generation Algorithm, Code-Generators.



Rayat Shikshan Sanstha's
Karmaveer Bhaurao Patil College of Engineering, Satara
Academic Year: 2025-26

Text Book:

1. Aho, Sethi, Ullman, Compilers Principles, Techniques and Tools, Pearson Education India, 2nd Edition, 2013

Reference Books:

1. Hopcroft, Motwani and Ullman, Introduction to Automata Theory, Languages and Computation, Pearson Publication, 2nd Edition, 2001.
2. Dick Grune, Kees van Reeuwijk, Henri E. Bal, Criel J. H. Jacobs and Koen Langendoen, Modern Compiler Design, Springer, 2nd Edition, 2012.

Course Coordinator

HOD